

CS & IT ENGINEERING



Computer Network

Basics of Switching



Lecture No. - 09



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Recap of Previous Lecture



Topic

Network Address Translation





Topics to be Covered



Topic

Basics of Switching





Topic : Syllabus



→ Basics of switching (circuit, packet and virtual circuit)

(→ CS-05)

and performance metrics;

V.V.Imp



Topic : Switching



→ Process to move data (or packets) towards destination over the network



Topic : Switching

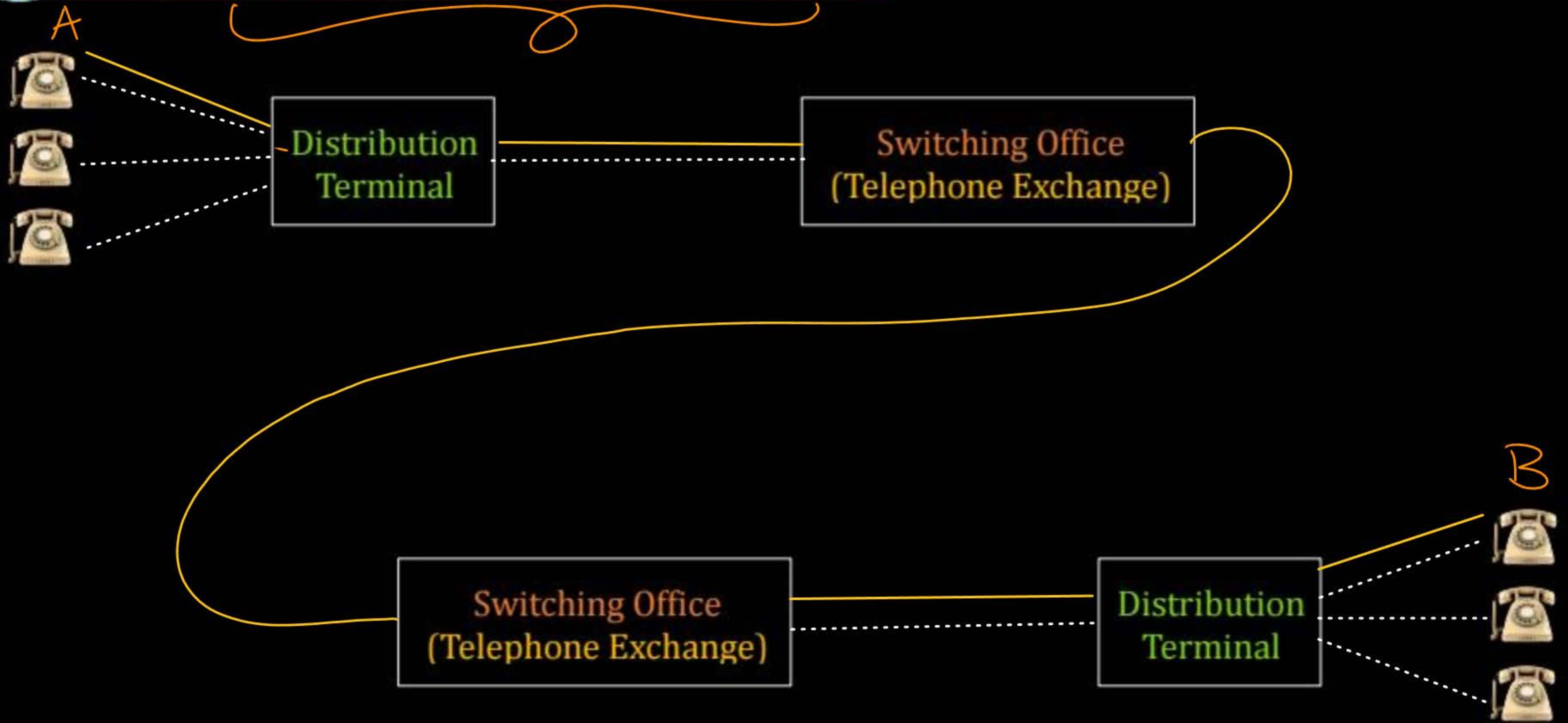


→ Types of switching techniques :

1. Circuit Switching
2. Packet Switching
3. Virtual Circuit Switching

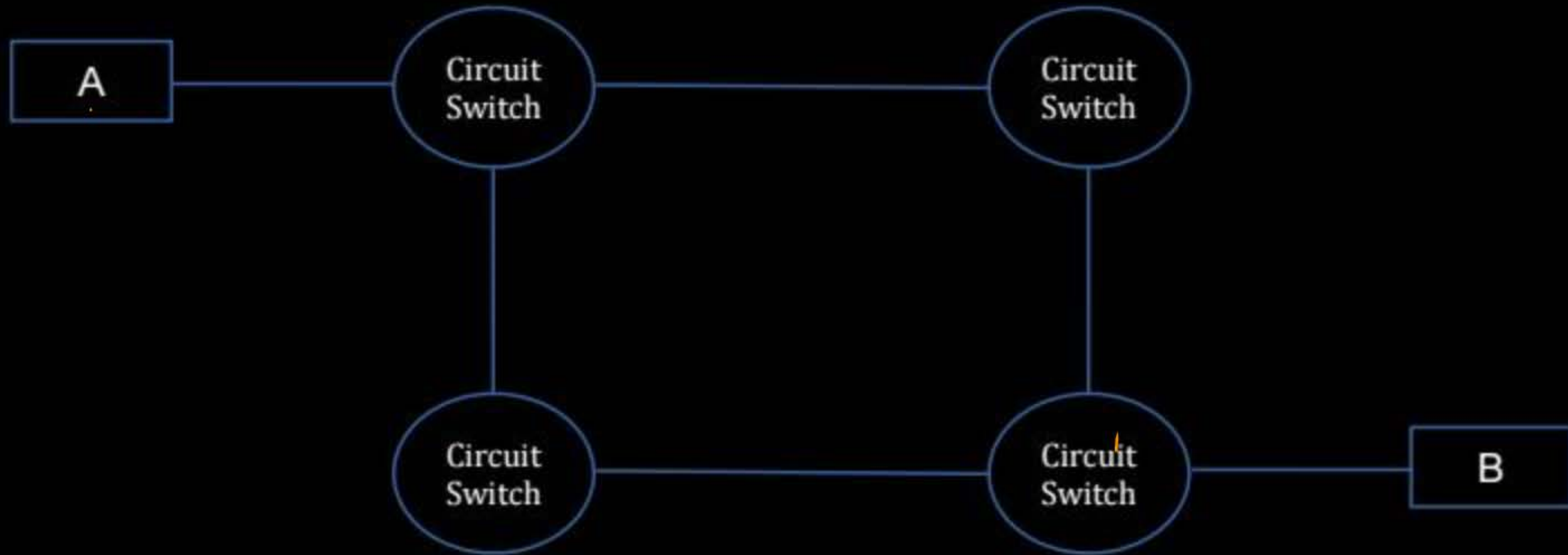


Topic : Circuit Switched Network





Topic : Circuit Switched Network





Topic : Circuit Switching



→ Establishes dedicated circuit between sender and receiver, before transmission

[Over the links of the network]

* path reservation



Topic : Circuit Switching



→ Example : Telephone Networks

[PSTN : Public Switched Telephone Network]



Topic : Circuit Switching



→ Phases of Circuit Switching :

1. Circuit establishment

✓ → path reserved

2. Data transfer

3. Circuit disconnect

→ Release the path



Topic : Circuit Switching



=> Circuit in a link implemented with

1. FDM

→ Analog (Frequency Division Mux)

2. TDM

→ Digital (Time Division Mux)



Topic : Circuit Switching



Advantage :-

→ No any congestion occur, during data transfer, over reserved path



Topic : Circuit Switching



Advantage :-

- All data (or packets) follow each other on reserved path
[Data (or packets) having same end-to-end delay]

↓
Due to no congestion
*very less variation



Topic : Circuit Switching



Disadvantage :-

→ Inefficient utilization of network resources





Topic : Circuit Switching



Disadvantage :-

→ Congestion may occur during circuit establishment



Topic : Circuit Switching



Disadvantage :-

→ Expensive



Topic : Computer Network

→ Hosts can be connected either directly
or indirectly through some networking devices

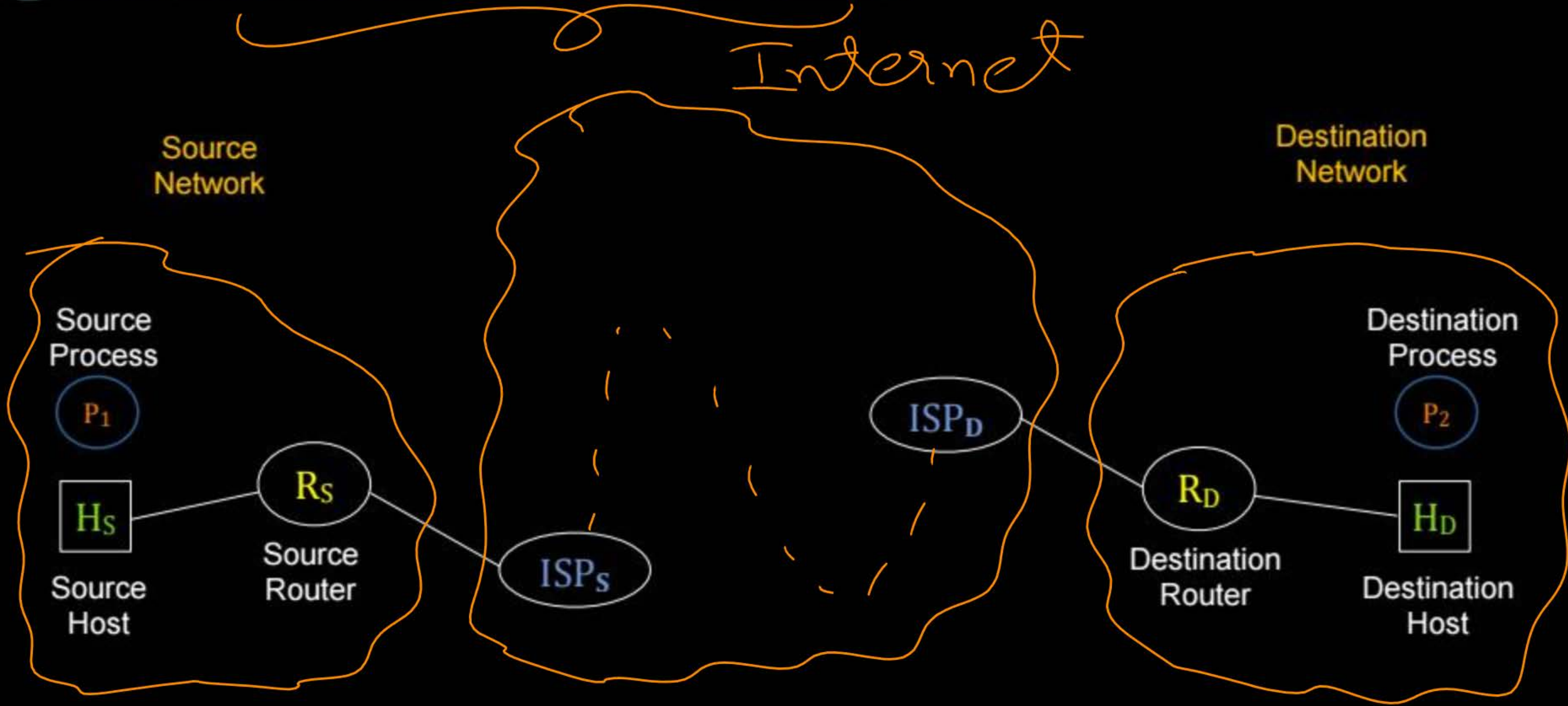
→ End devices.

→ Host : Computer, Server or Smartphone

→ Networking devices : Switch, Router etc



Topic : Packet Switched Network





Topic : Two Process Communication

→ Two network processes, belongs to different hosts

Communicate over network, by exchanging "messages"



Topic : Message Switching



→ No any **established circuit** required between **sender** and **receiver**

→ **Store and Forward Network**



Topic : Message Switching



→ Entire message is transmitted as single unit



Topic : Packet Switching



→ Message is divided into smaller size packets

[Packets may be same or different size]

→ IP Packets



Topic : Packet Switching



* No path reservation before transmission.

→ No any **established circuit** required between **sender** and **receiver**

→ **Store and Forward** [**Datagram Network**] → Packet switched network

↓
TCP/IP

↓
OSI



Topic : Packet Switching

⇒ TCP/IP Model
⇒ OSI Model



→ Example : Internet is a packet-switched network

→ Every WAN
→ largest WAN



Topic : Packet Switching



→ Every packet is treated independently at every intermediate router



Topic : Packet Switching



Advantage :-

→ Efficient utilization of network resources
[Lead to better utilization of bandwidth resource]

→ Less Expensive



Topic : Packet Switching



Disadvantage :-

→ Congestion may occur during routing





Topic : Packet Switching



Disadvantage :-

=> **More** per packet **processing overhead** at intermediate router

↓
is part of store and forward delay



Topic : Packet Switching



Disadvantage :-

=> Packets may follow different routing paths



Topic : Packet Switching



Disadvantage :-

=> Packets may have different end-to-end delay

[MCQ] [GATE-CS-2005] [1-Mark] IIT-B



#Q. Which one of the following statements is FALSE?

- A Packet switching leads to better utilization of bandwidth resources than circuit switching TRUE
- ✓ B Packet switching results in less variation in delay than circuit switching False
→ More
- C Packet switching requires more per-packet processing than circuit switching TRUE
→ store & forward
- D Packet switching can lead to reordering unlike in circuit switching True

Ans: B



Topic : Virtual Circuit Switching

→ Need to establish virtual circuit
between sender and receiver before transmission

- * path selection over packet switched network.
- * No path reservation physically



Topic : Virtual Circuit Switching

→ 3 phase in virtual circuit switching :

1. VC setup → IP protocol used for VC setup only
2. Data Transfer → No IP Protocol
3. VC teardown



Topic : Virtual Circuit Switching



- Entire routing path of packets is determined before transmission
[Entire routing path is fixed for duration of virtual circuit]



Topic : Virtual Circuit Switching



→ Example : Used in technologies like X.25, Frame Relay, and ATM



Topic : Virtual Circuit Switching



Advantage :-

→ Every **packets** follow each other on **predefined path**

* No reordering is required at destination host.



Topic : Virtual Circuit Switching

Disadvantage :-

- Congestion may occur during routing
- Packets may have different end-to-end delay



2 mins Summary



Topic

Basics of Switching

THANK - YOU